

Task 2: Multiple Sequence Alignment (MSA)

Bioinformatics Internship Report — CodeAlpha

1. Introduction

Multiple Sequence Alignment (MSA) is a fundamental approach in computational biology used to align three or more biological sequences of similar length. By aligning these sequences simultaneously, researchers can infer evolutionary history, identify highly conserved structures, and discover functional motifs that are critically required for the survival and activity of a specific protein family.

In this task, the large subunit chain of the enzyme **Ribulose biphosphate carboxylase oxygenase (Rubisco)** was selected. Rubisco acts as the primary molecular engine driving global biological carbon fixation during the photosynthetic pathway. Analyzing this baseline metabolic catalyst across highly distinct plant families reveals structural constraints and pinpointed conservation essential for survival.

2. Methodology

Five homologous sequences of the Rubisco large subunit were retrieved from the UniProt database to map multi-species sequence alignment configurations:

Organism	Scientific Name	Common Name	Database Entry Reference
	<i>Arabidopsis thaliana</i>	Thale Cress (Model Plant)	UniProtKB - O03042 (RBL_ARATH)
	<i>Solanum lycopersicum</i>	Tomato	UniProtKB Homolog / BLAST Search Hit
	<i>Oryza sativa</i>	Rice	UniProtKB Homolog / BLAST Search Hit
	<i>Triticum aestivum</i>	Wheat	UniProtKB Homolog / BLAST Search Hit
	<i>Vitis vinifera</i>	Grape	UniProtKB Homolog / BLAST Search Hit

- **Platform Suite:** EMBL-EBI JDispatcher Analysis Interface.
- **Core Alignment Algorithm:** Clustal Omega, utilizing progressive alignment profiles guided by seeded trees to calculate precise positional matching across varying sequences.

- **Visualization Adjustments:** The alignment output was color-coded using standard biochemical properties to visually accentuate variations and conserved residues.

3. Results and Structural Interpretation

The job submission was successfully processed through the Clustal Omega algorithm pipeline. Below is the multi-sequence data submission layout captured directly during the run execution environment:

The screenshot shows the 'Submit a Clustal Omega Job' form. The title is 'Multiple Sequence Alignment'. The sequence type is 'Protein'. The input sequence is a multi-FASTA format containing five entries: *Triticum aestivum*, *Vitis vinifera*, *Arabidopsis thaliana*, *Solanum lycopersicum*, and *Oryza sativa*. The interface includes a 'Use a Sample Sequence' button and a note about providing sequences by typing or uploading a file.

Figure 1: Clustal Omega multi-FASTA submission

Upon full execution, the generated structural profile demonstrates an exceptionally tight sequence similarity among all five botanical entries. The complete structural output map and alignment overview indicate optimal evolutionary packing:

The screenshot shows the 'Output' section of the Clustal Omega analysis summary. It displays the job ID: clustalo-I20260521-021317-0764-53948670-p2m. Below the job ID, there are links for 'Output', 'Guide Tree', 'Phylogenetic Tree', 'Download', and 'Submission'. The main content is the 'CLUSTAL O(1.2.4) multiple sequence alignment' output, which shows the alignment of the five sequences. The alignment is color-coded to highlight conserved residues and variations. The sequences are: *Arabidopsis_thaliana_Rubisco*, *Solanum_lycopersicum_Tomato_Rubisco*, *Vitis_vinifera_Grape_Rubisco*, *Oryza_sativa_Rice_Rubisco*, and *Triticum_aestivum_Wheat_Rubisco*. The alignment shows a high degree of similarity, with many identical residues across all five sequences.

Figure 2: Clustal Omega job configuration parameters and analysis summary window.

The precise residue-by-residue layout visualization confirms that the vast majority of amino acid blocks are strictly identical. This structural uniformity is visible through the color blocks and consensus scoring markers below:

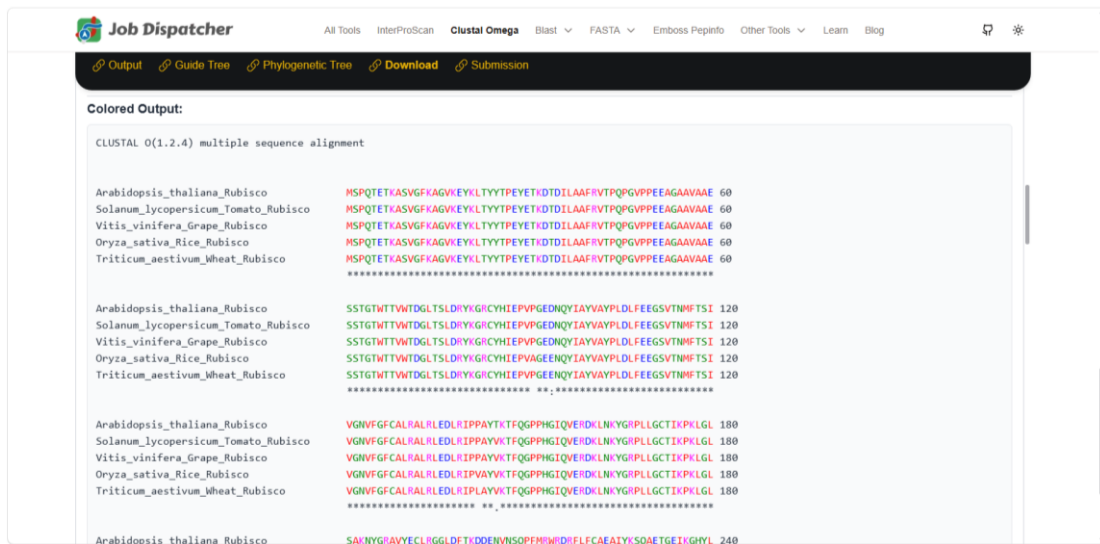


Figure 3: Color-coded residue alignment segments highlighting massive sequence identity blocks.

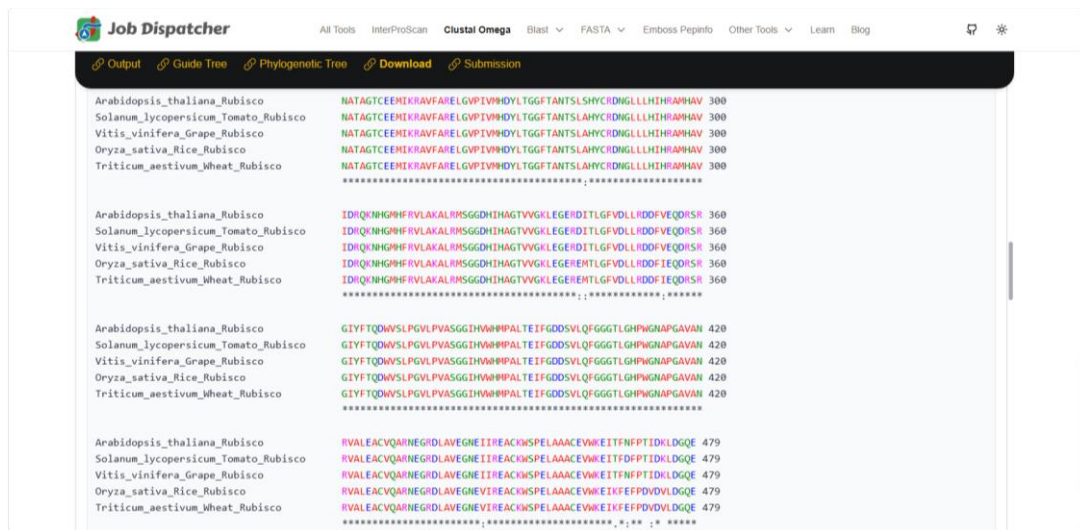


Figure 4: Continuous alignment sequences showcasing the invariant consensus string positions.

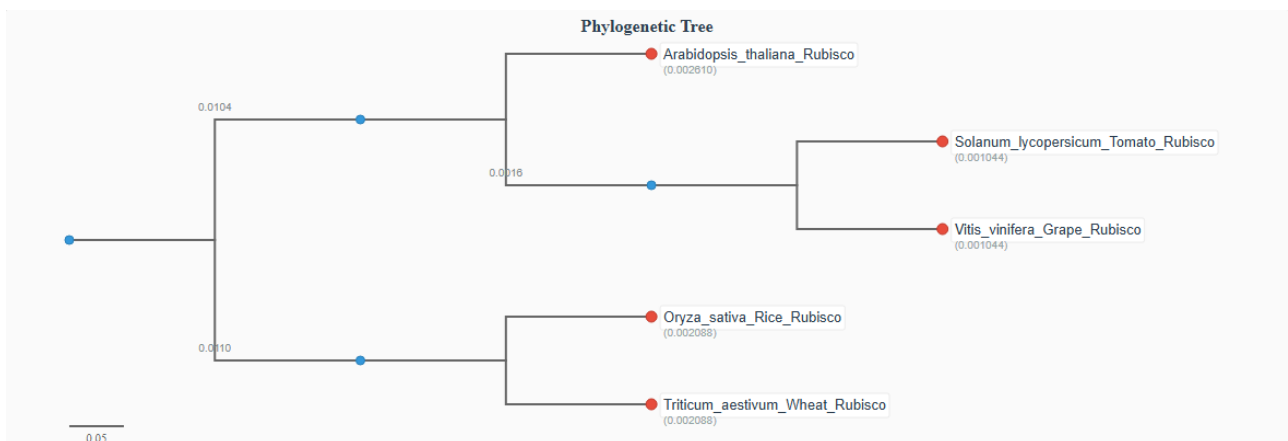


Figure 5: Phylogenetic Tree

Consensus Sequence Scoring Key:

- Asterisk (*):** Denotes an amino acid position that is completely unchanged and 100% identical across all 5 organisms.
- Colon (:):** Demonstrates conservation between residues possessing strongly similar biochemical characteristics.

- **Period (.)**: Demonstrates partial conservation between residues with weakly similar chemical traits.

The profound presence of continuous consensus asterisks (*) separating monocots (Rice, Wheat) from dicots (Tomato, Grape, Arabidopsis) shows that the catalytic active pocket required to accept RuBP and capture carbon dioxide has been locked under absolute evolutionary selection. Non-disruptive conservative mutations (indicated by colons) are minor and restricted strictly to surface-exposed regions.

4. Conclusion

The Clustal Omega MSA successfully verified the evolutionary rigid layout of the Rubisco large subunit chain across multiple plant lineages. The absolute sequence identity observed at the functional zones serves as definitive evidence of strict evolutionary constraints. These mapped invariant core motifs provide a highly detailed molecular blueprint for potential genetic engineering frameworks aiming to optimize photosynthetic turnover and crop yields.